

SuperBundle - Bundle Adjustment with Superquadric Primitives

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Abstract

The ABSTRACT is to be in fully justified italicized text, at the top of the left-hand column, below the author and affiliation information. Use the word “Abstract” as the title, in 12-point Times, boldface type, centered relative to the column, initially capitalized. The abstract is to be in 10-point, single-spaced type. Leave two blank lines after the Abstract, then begin the main text. Look at previous 3DV abstracts to get a feel for style and length.

1. Introduction

Reconstructing 3D scenes from multiple camera images is a classical problem in Computer Vision. There are many feed-forward models for predicting camera parameters and point clouds to reconstruct full 3D scenes [9, 18, 20]. However, especially in sparse view settings, camera pose estimates can drift or be imprecise. This comes from the limited overlap of the views, ambiguous geometry, and scale uncertainty. To reduce the prediction error one can use optimization based methods like bundle adjustment [17] to refine the reconstructed scenes.

In this project, we improved the camera poses in very sparse view settings with a modified version of the classic bundle adjustment algorithm. We used a superquadric [14] representation of our 3D scene as a structural prior, improving the camera pose estimates compared to the classic bundle adjustment. For this, we used the superquadric representation of the Aria Synthetic Environments Dataset[5] generated with Superdec [7] from the ground truth point clouds.

XXX Add stuff with connection to results and discussion

2. Related Works

Feed Forward Reconstruction Traditional multi-view reconstruction pipelines [8, 13, 15] rely on sequential stages of feature extraction, matching, and geometric optimization, making them computationally expensive and sensitive to initialization. Feed-forward reconstruction models address this by directly regressing 3D structure and camera parameters from images in a single forward pass. Feed-forward reconstruction models address this by directly regressing 3D structure and camera parameters from images in a single forward pass. DUST3R [19] and MAST3R [12] established this paradigm by predicting dense point maps and relative camera poses from image pairs without requiring known intrinsics, relying on post-hoc global alignment to extend beyond pairs. Methods like VGGT [18] or MapAnything [9] advanced this further by processing up to hundreds of images simultaneously in a single forward pass, jointly predicting camera parameters, depth maps, point maps, and 3D point tracks.

Bundle Adjustment Bundle adjustment (BA) is the gold standard for refining camera poses and 3D structure in multi-view reconstruction. Formally introduced by Triggs et al. [17], it minimizes the total reprojection error over all cameras and scene points simultaneously via the Levenberg-Marquardt algorithm, exploiting the sparse block structure of the Jacobian to keep the problem tractable. In classical SfM pipelines such as COLMAP [15], BA is applied both locally after each image registration and globally as a final refinement, preventing drift and enforcing geometric consistency across the full reconstruction. While highly effective given sufficient image overlap and reliable correspondences, classical BA is sensitive to initialization and provides no mechanism to recover from fundamentally incorrect pose estimates. This limitation becomes particularly acute in the sparse-view setting.

Superquadrics Superquadrics are a family of parametric geometric primitives, first introduced by Barr [6] and later popularized for 3D shape recovery by Solina and Bajcsy [16], capable of representing a wide range of shapes — from boxes to cylinders to ellipsoids — with a small set of parameters encoding position, orientation, scale, and two shape exponents. Their compactness and analytical differentiability make them attractive as structural scene representations. SuperDec [7] builds on this foundation by presenting an approach for creating compact 3D scene representations through decomposition into superquadric primitives, designing a novel architecture that efficiently decomposes point clouds of arbitrary objects into a compact set of superquadrics. Rather than targeting photorealistic reconstruction, SuperDec solves the decomposition locally on individual objects and leverages instance segmentation to scale to full 3D scenes. In our work, we use SuperDec to obtain a compact geometric abstraction of the scene from the ground truth point cloud, which we then use as a structural prior to constrain and guide bundle adjustment optimization.

3. Method

4. Results

5. Discussion

6. Introduction

Please follow the steps outlined below when submitting your manuscript to the IEEE Computer Society Press. This style guide now has several important modifications (for example, you are no longer warned against the use of sticky tape to attach your artwork to the paper), so all authors should read this new version.

6.1. Language

All manuscripts must be in English.

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Please refer to the author guidelines on the 3DV 2026 web page for a discussion of the policy on dual submissions.

6.3. Paper length

Papers, excluding the references section, must be no longer than eight pages in length. The references section will not be included in the page count, and there is no limit on the length of the references section. For example, a paper of eight pages with two pages of references would have a total length of 10 pages. **There will be no extra page charges for 3DV 2026.**

Overlength papers will simply not be reviewed. This includes papers where the margins and formatting are deemed

to have been significantly altered from those laid down by this style guide. Note that this L^AT_EX guide already sets figure captions and references in a smaller font. The reason such papers will not be reviewed is that there is no provision for supervised revisions of manuscripts. The reviewing process cannot determine the suitability of the paper for presentation in eight pages if it is reviewed in eleven.

6.4. The ruler

The L^AT_EX style defines a printed ruler which should be present in the version submitted for review. The ruler is provided in order that reviewers may comment on particular lines in the paper without circumlocution. If you are preparing a document using a non-L^AT_EX document preparation system, please arrange for an equivalent ruler to appear on the final output pages. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera-ready copy should not contain a ruler. (L^AT_EX users may use options of `cvpr.sty` to switch between different versions.)

Reviewers: note that the ruler measurements do not align well with lines in the paper — this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. Just use fractional references (*e.g.*, this line is 087.5), although in most cases one would expect that the approximate location will be adequate.

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Please number all of your sections and displayed equations as in these examples:

$$E = m \cdot c^2 \tag{1}$$

and

$$v = a \cdot t. \tag{2}$$

It is important for readers to be able to refer to any particular equation. Just because you did not refer to it in the text does not mean some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like “the equation second from the top of page 3 column 1”. (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin’s description of how to write mathematics: <http://www.pamitc.org/documents/mermin.pdf>.

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Many authors misunderstand the concept of anonymizing for blind review. Blind review does not mean that one must remove citations to one's own work—in fact it is often impossible to review a paper unless the previous citations are known and available.

Blind review means that you do not use the words “my” or “our” when citing previous work. That is all. (But see below for tech reports.)

Saying “this builds on the work of Lucy Smith [1]” does not say that you are Lucy Smith; it says that you are building on her work. If you are Smith and Jones, do not say “as we show in [7]”, say “as Smith and Jones show in [7]” and at the end of the paper, include reference 7 as you would any other cited work.

An example of a bad paper just asking to be rejected:

An analysis of the frobnicatable foo filter.

In this paper we present a performance analysis of our previous paper [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.

[1] Removed for blind review

An example of an acceptable paper:

An analysis of the frobnicatable foo filter.

In this paper we present a performance analysis of the paper of Smith *et al.* [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.

[1] Smith, L and Jones, C. “The frobnicatable foo filter, a fundamental contribution to human knowledge”. *Nature* 381(12), 1-213.

If you are making a submission to another conference at the same time, which covers similar or overlapping material, you may need to refer to that submission in order to explain the differences, just as you would if you had previously published related work. In such cases, include the anonymized parallel submission [10] as supplemental material and cite it as

[1] Authors. “The frobnicatable foo filter”, F&G 2014 Submission ID 324, Supplied as supplemental material `fg324.pdf`.

Finally, you may feel you need to tell the reader that more details can be found elsewhere, and refer them to a technical report. For conference submissions, the paper must stand on its own, and not *require* the reviewer to go to a tech report for further details. Thus, you may

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Sometimes your paper is about a problem which you tested using a tool that is widely known to be restricted to a single institution. For example, let's say it's 1969, you have solved a key problem on the Apollo lander, and you believe that the 1970 audience would like to hear about your solution. The work is a development of your celebrated 1968 paper entitled “Zero-g frobnication: How being the only people in the world with access to the Apollo lander source code makes us a wow at parties”, by Zeus *et al.*

You can handle this paper like any other. Do not write “We show how to improve our previous work [Anonymous, 1968]. This time we tested the algorithm on a lunar lander [name of lander removed for blind review]”. That would be silly, and would immediately identify the authors. Instead write the following:

We describe a system for zero-g frobnication. This system is new because it handles the following cases: A, B. Previous systems [Zeus *et al.* 1968] did not handle case B properly. Ours handles it by including a foo term in the bar integral.

...

The proposed system was integrated with the Apollo lunar lander, and went all the way to the moon, don't you know. It displayed the following behaviours, which show how well we solved cases A and B: ...

As you can see, the above text follows standard scientific convention, reads better than the first version, and does not explicitly name you as the authors. A reviewer might think it likely that the new paper was written by Zeus *et al.*, but cannot make any decision based on that guess. He or she would have to be sure that no other authors could have been contracted to solve problem B.

FAQ

Q: Are acknowledgements OK?

A: No. Leave them for the final copy.

Q: How do I cite my results reported in open challenges?

A: To conform with the double-blind review policy, you can report results of other challenge participants together with your results in your paper. For your results, however, you should not identify yourself and should not mention your participation in the challenge. Instead present your results referring to the method proposed in your paper and draw conclusions based on the experimental comparison to other results.

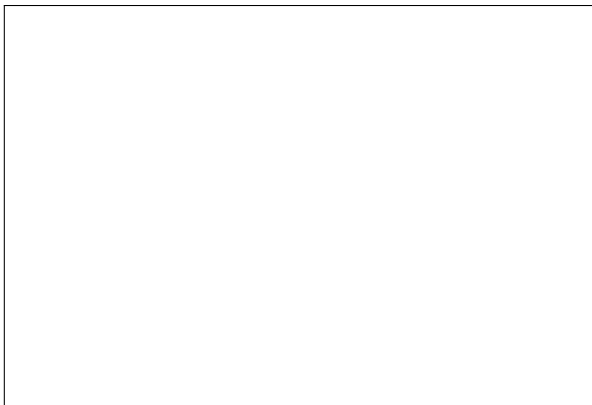


Figure 1. Example of caption. It is set in Roman so that mathematics (always set in Roman: $B \sin A = A \sin B$) may be included without an ugly clash.

6.8. Miscellaneous

Compare the following:

```
$conf_a$          confa
$\mathit{conf}_a$  confa
```

See The $\text{\TeX}$ book, p165.

The space after *e.g.*, meaning “for example”, should not be a sentence-ending space. So *e.g.* is correct, *e.g.* is not. The provided `\eg` macro takes care of this.

When citing a multi-author paper, you may save space by using “et alia”, shortened to “*et al.*” (not “*et. al.*” as “*et*” is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al.* However, use it only when there are three or more authors. Thus, the following is correct: “Frobnication has been trendy lately. It was introduced by Alpher [1], and subsequently developed by Alpher and Fotheringham-Smythe [2], and Alpher *et al.* [4].”

This is incorrect: “... subsequently developed by Alpher *et al.* [2] ...” because reference [2] has just two authors.

7. Formatting your paper

All text must be in a two-column format. The total allowable size of the text area is $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. Columns are to be $3\frac{1}{4}$ inches (8.25 cm) wide, with a $\frac{5}{16}$ inch (0.8 cm) space between them. The main title (on the first page) should begin 1 inch (2.54 cm) from the top edge of the page. The second and following pages should begin 1 inch (2.54 cm) from the top edge. On all pages, the bottom margin should be $1\frac{1}{8}$ inches (2.86 cm) from the bottom edge of the page for 8.5×11 -inch paper; for A4 paper, approximately $1\frac{5}{8}$ inches (4.13 cm) from the bottom edge of the page.

7.1. Margins and page numbering

All printed material, including text, illustrations, and charts, must be kept within a print area $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. Page numbers should be in the footer, centered and $\frac{3}{4}$ inches from the bottom of the page. The review version should have page numbers, yet the final version submitted as camera ready should not show any page numbers. The $\text{\LaTeX}$ template takes care of this when used properly.

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Wherever Times is specified, Times Roman may also be used. If neither is available on your word processor, please use the font closest in appearance to Times to which you have access.

MAIN TITLE. Center the title $1\frac{3}{8}$ inches (3.49 cm) from the top edge of the first page. The title should be in Times 14-point, boldface type. Capitalize the first letter of nouns, pronouns, verbs, adjectives, and adverbs; do not capitalize articles, coordinate conjunctions, or prepositions (unless the title begins with such a word). Leave two blank lines after the title.

AUTHOR NAME(s) and AFFILIATION(s) are to be centered beneath the title and printed in Times 12-point, non-boldface type. This information is to be followed by two blank lines.

The **ABSTRACT** and **MAIN TEXT** are to be in a two-column format.

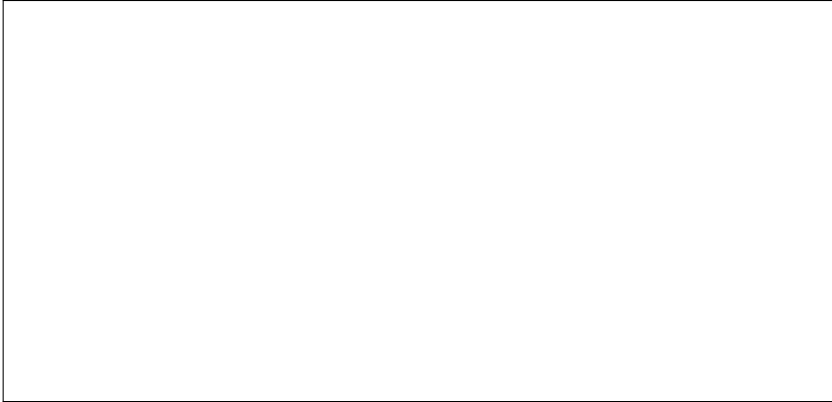
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Figure and table captions should be 9-point Roman type as in Figs. 1 and 2. Short captions should be centred.

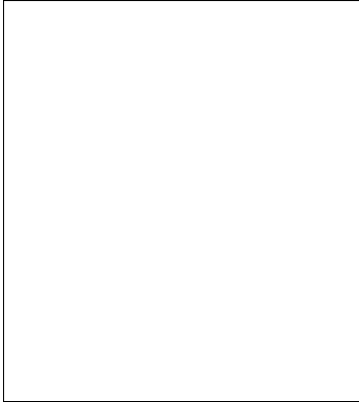
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FIRST-ORDER HEADINGS. (For example, **1. Introduction**) should be Times 12-point boldface, initially capitalized, flush left, with one blank line before, and one blank line after.

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(a) An example of a subfigure.



(b) Another example of a subfigure.

Figure 2. Example of a short caption, which should be centered.

7.3. Footnotes

Please use footnotes¹ sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the bottom of the column on the page on which it is referenced. Use Times 8-point type, single-spaced.

7.4. Cross-references

For the benefit of author(s) and readers, please use the

`\cref{...}`
command for cross-referencing to figures, tables, equations, or sections. This will automatically insert the appropriate label alongside the cross-reference as in this example:

To see how our method outperforms previous work, please see Fig. 1 and Tab. 1. It is also possible to refer to multiple targets as once, *e.g.* to Figs. 1 and 2a. You may also return to Sec. 7 or look at Eq. (2).

If you do not wish to abbreviate the label, for example at the beginning of the sentence, you can use the

`\Cref{...}`
command. Here is an example:

Figure 1 is also quite important.

7.5. References

List and number all bibliographical references in 9-point Times, single-spaced, at the end of your paper. When referenced in the text, enclose the citation number in square

¹This is what a footnote looks like. It often distracts the reader from the main flow of the argument.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one’s heart Frob

Table 1. Results. Ours is better.

brackets, for example [10]. Where appropriate, include page numbers and the name(s) of editors of referenced books. When you cite multiple papers at once, please make sure that you cite them in numerical order like this [1–3, 10, 11]. If you use the template as advised, this will be taken care of automatically.

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All graphics should be centered. In L^AT_EX, avoid using the `center` environment for this purpose, as this adds potentially unwanted whitespace. Instead use

`\centering`

at the beginning of your figure. Please ensure that any point you wish to make is resolvable in a printed copy of the paper. Resize fonts in figures to match the font in the body text, and choose line widths that render effectively in print. Readers (and reviewers), even of an electronic copy, may choose to print your paper in order to read it. You cannot insist that they do otherwise, and therefore must not assume that they can zoom in to see tiny details on a graphic.

When placing figures in L^AT_EX, it’s almost always best to use `\includegraphics`, and to specify the figure width as a multiple of the line width as in the example below

```
\usepackage{graphicx} ...
\includegraphics[width=0.8\linewidth]
{myfile.pdf}
```

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Supplementary Material

9. Rationale

Having the supplementary compiled together with the main paper means that:

- The supplementary can back-reference sections of the main paper, for example, we can refer to Sec. 6;
- The main paper can forward reference sub-sections within the supplementary explicitly (e.g. referring to a particular experiment);
- When submitted to arXiv, the supplementary will already included at the end of the paper.

To split the supplementary pages from the main paper, you can use [Preview \(on macOS\)](#), [Adobe Acrobat](#) (on all OSs), as well as [command line tools](#).